Retrieval Degradation in Multi-Turn Agentic RAG Systems: Observations from a Production Deployment

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Abstract

Retrieval-Augmented Generation (RAG) has demonstrated strong performance across single-turn question-answering benchmarks. Production agentic RAG systems, however, decompose a problem into sub-queries and retrieve once per reasoning turn, and existing evaluations do not examine how retrieval behaves across these turns. We study *multi-turn retrieval degradation*, which we define as a decline in retrieval similarity across the sequential reasoning turns of a session. Using instrumented logs of 150 sessions (550 turns) from *Math Professor AI*, a publicly deployed mathematics tutoring system with a TF-IDF retriever, we find that mean top-chunk similarity declines modestly across turns (-0.0075 per turn, $p = 0.004$, session-clustered; a 6.6% decline between turn 1 and turn 4, $p = 0.046$) and is moderately correlated with the length of the accumulated reasoning context (Pearson $r = -0.283$, $p < 0.0001$). In this system the retrieval query contains only the original question and the current sub-query, so the decline is not an effect of context on the retriever; it must come from the sub-queries, the index, or the sessions that reach later steps, and within sessions context length and turn number cannot be separated. An exploratory estimator and a context-length baseline reach AUROC 0.63–0.70 at predicting final-answer feedback on 102 labeled sessions, but with only 8 negative labels all confidence intervals include chance. We discuss what the logs can and cannot show and which controlled experiments would separate the candidate explanations.

1. Introduction

Retrieval-Augmented Generation (RAG) addresses a fundamental limitation of large language models: the inability to reliably ground outputs in external, up-to-date knowledge (Lewis et al., 2020; Karpukhin et al., 2020). The dominant paradigm assumes a single-turn interaction: one query, one retrieval, one generation.

The proliferation of agentic LLM systems challenges this assumption. In agentic settings, an LLM plans a sequence of sub-queries, retrieves context for each, generates intermediate reasoning steps conditioned on accumulated context, and synthesises a final response. This architecture — instantiated in systems like ReAct (Yao et al., 2023), Self-RAG (Asai et al., 2023), and FLARE (Jiang et al., 2023) — makes retrieval a repeated operation within a single session. The later steps of a session differ from the first: the plan has progressed and the accumulated reasoning has grown.

Multi-turn conversational benchmarks such as TopiOCQA (Adlakha et al., 2022) and MTRAG (Katsis et al., 2025) study retrieval across the turns of a user conversation, but few studies measure how retrieval quality evolves across the internal reasoning turns of a deployed agentic system. We use the term *multi-turn retrieval degradation* for a decline in retrieval similarity across these

turns and present an empirical study grounded in *Math Professor AI*, a production-deployed agentic RAG system for mathematics tutoring (`math-professor-agent-k6h4.vercel.app`). Our main contributions are:

- Empirical documentation, across 150 sessions (550 turn observations), of a modest decline in top-chunk similarity per turn ($p = 0.004$, session-clustered) and of a moderate correlation between similarity and accumulated reasoning length ($r = -0.283$, $p < 0.0001$).
- An analysis of what these associations can support: because the retrieval query never contains accumulated context, and context length and turn number are inseparable within sessions, we attribute the decline to the sub-queries, the index or the sessions that reach later steps rather than to context interference, and list candidate explanations and tests.
- An exploratory turn-aware confidence estimator (AUROC = 0.634) for predicting final-answer feedback; context length alone reaches AUROC = 0.695, though all intervals include chance given only 8 negative labels.
- Implications for RAG evaluation, including what to log. Query anchoring, which we describe precisely, is a design feature whose effect we could not test, since it was enabled in every logged turn.

2. Background and Related Work

2.1 Retrieval-Augmented Generation

The original RAG framework (Lewis et al., 2020) introduced retrieve-then-generate architecture. Subsequent work extended RAG to open-domain QA, knowledge-intensive tasks, and multi-hop reasoning. Much of this work evaluates retrieval in single-turn settings: one query, one retrieval, assessed independently of prior context.

2.2 Iterative and Agentic RAG

Recent systems have moved beyond single-turn retrieval. FLARE (Jiang et al., 2023) monitors generation confidence and triggers additional retrievals. Corrective RAG (Yan et al., 2024) evaluates retrieved document quality and triggers corrective web search. Self-RAG (Asai et al., 2023) trains the LLM to decide when to retrieve. IterKey (Hayashi et al., 2025) demonstrates that iteratively refining retrieval keywords within a single reasoning context significantly improves output quality. These works are evaluated by end-task quality rather than by per-turn retrieval similarity. Multi-turn conversational benchmarks such as TopiOCQA (Adlakha et al., 2022) and MTRAG (Katsis et al., 2025) study retrieval across the turns of a user conversation; our turns are instead internal sub-queries issued while solving a single problem.

2.3 Long Context and Retrieval

Liu et al. (2023) demonstrate that LLM performance degrades for relevant documents placed in the middle of long contexts — the "lost in the middle" phenomenon. Ma and Okazaki (2026) identify specific attention heads in long-context LLMs that specialise in retrieval, showing these retrieval heads can be optimised to improve long-context performance. Our observations are not inconsistent with such effects, but our retriever is a TF-IDF index outside the LLM, so we cannot test them. Such effects could arise in systems whose retrieval queries are conditioned on accumulated context; ours are not.

2.4 Confidence Estimation

Ozaki et al. (2025) study retrieval confidence calibration in medical RAG systems, finding that raw confidence scores do not reliably predict answer correctness in single-turn settings. Yoshikawa and Okazaki (2023) introduce selective prediction for language models. Our turn-aware estimator extends these ideas to the multi-turn agentic setting.

3. System Description: Math Professor AI

3.1 Architecture Overview

Math Professor AI is a publicly deployed agentic RAG system for step-by-step mathematics tutoring. Gemini 2.5 Flash handles input guardrail validation and problem-type classification; Gemini 2.5 Pro decomposes the question, generates each reasoning step and synthesises the final answer.

Component	Technology	Function
Input Guardrail	Gemini 2.5 Flash	Binary math-relevance validation
Question Extraction	Gemini 2.5 Flash	Extract questions from documents
Semantic Retrieval	TF-IDF + Cosine Sim.	Document chunk retrieval (top-3)
KB Fallback	Keyword Matching	Pre-stored answer lookup
Web Grounding	Google Search API	Real-time web retrieval
Solution Engine	Gemini 2.5 Pro	Step-by-step answer generation
Answer Refinement	Gemini 2.5 Pro	Refine answers from human feedback
Feedback Loop	React / TypeScript	Human correction collection

Table 1: System components and technologies.

3.2 Retrieval Pipeline Detail

Documents are chunked by a Python service that splits text into sentences, detects topic shifts with a sentence-transformer, groups sentences into units of at most 512 tokens, and adds an overlap of 64 tokens between chunks. Chunks are stored in the browser's localStorage. TF-IDF vectors are computed at retrieval time from all stored chunks: the vocabulary is the (up to) 10,000 terms with the highest document frequency, term frequency is sublinear ($1 + \log$ of the count), inverse document frequency is smoothed, and vectors are L2-normalised, so cosine similarity is a dot product. The system returns the top three chunks whose similarity to the query is at least 0.15. If no chunk passes the threshold, it falls back to keyword matching against a small built-in knowledge base (Tier 2) and then to Google Search grounding (Tier 3). In our data collection the index was built from a researcher-prepared text file of math questions; the system can also re-index refined answers entered through its feedback interface.

3.3 Agentic Reasoning Loop

For each user question the system (1) asks Gemini 2.5 Pro once to decompose the question into 2–5 sub-queries $\{q_1, q_2, \ldots, q_n\}$, falling back to the original question as a single sub-query if decomposition fails. Then, for each q_i in order, it (2) retrieves context C_i, and (3) generates an intermediate reasoning step r_i conditioned on C_i, the original question and all prior steps. Finally it (4) synthesises a final answer from $\{r_1, \ldots, r_n\}$. The accumulated reasoning therefore grows with every turn, but it is used only for generation. The retrieval query for turn i is the original question followed by q_i (**query anchoring**); accumulated reasoning is never part of the retrieval query, and the sub-queries are produced before any retrieval, so they do not depend on it either.

Query anchoring was enabled in all logged turns. The system has a switch to retrieve with q_i alone, but we have no sessions with it off and could not evaluate its effect (Section 7).

3.4 Data Collection and Logging

Each turn is logged with: the sub-query and original question text, the number of retrieved chunks, the number of web-grounding chunks, a context token count, the retrieval tier, the TF-IDF cosine similarity of the top retrieved chunk (0 when no chunk passes the threshold), whether web search was used, whether anchoring was enabled, the problem type, and a correctness label when the user gave feedback. The context token count is the approximate size (characters divided by four) of the accumulated sub-queries and reasoning at the end of the turn, including the current step's output. It measures reasoning progress; it is not what the retriever receives. A correctness label is the user's feedback on the session's final answer and is stored on the last turn of the session.

We collected 150 sessions yielding 550 turn-level observations across calculus (43%), algebra (35%), geometry (11%), and statistics (10%) domains. Of the 550 turns, 547 used Tier 1 TF-IDF retrieval and 3 fell through to Tier 2 (similarity recorded as 0). Correctness labels were available for 102 of the 150 sessions (68%) — 94 correct and 8 incorrect — each attached to the last turn of its session; 78 of the 102 labeled sessions end at turn 4. These labels support a preliminary AUROC calculation. Sessions were collected at a coaching center from adult (18+) users who were aware that their use of the app was being logged for research. Logs contain only question text, session identifiers and system metadata, with no names or personal information, and all session data was fully anonymised prior to analysis.

4. Empirical Observations

4.1 Overview of Session Corpus

Our 150 sessions contained between 1 and 5 sub-query turns each, with a mean of 3.67 turns per session. The mean context length at turn 1 was 482 approximate tokens, growing to a mean of 2,095 at turn 4 — a 4.3× increase.

4.2 Context Length Growth Across Turns

Figure 1 shows mean context token count across reasoning turns 1 through 5. Context (the accumulated sub-queries and reasoning, in approximate tokens) grows monotonically and approximately linearly: 482 tokens at turn 1, 972 at turn 2, 1,464 at turn 3, 2,095 at turn 4, and 3,611 at turn 5. The shaded region represents ±1 standard deviation, showing that context growth

variance also increases at higher turns — consistent with more diverse problem complexity in sessions that extend to later turns.

Figure 1: Context length growth across turns

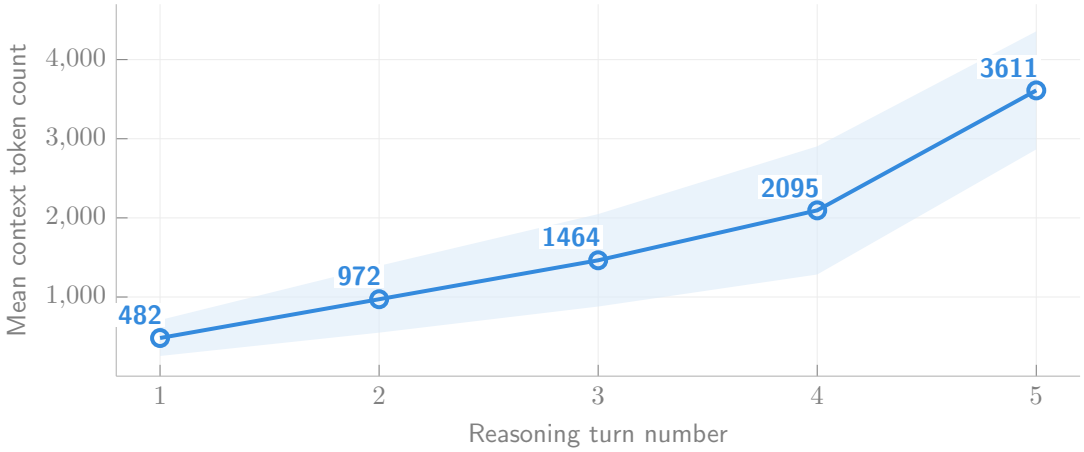


Figure 1: Mean context token count (approximate tokens of accumulated sub-queries and reasoning) by reasoning turn, from 482 at turn 1 to 2,095 at turn 4 — a 4.3× increase. Shaded region = ±1 standard deviation across sessions; turn 5 has only 3 sessions.

4.3 Retrieval Similarity Across Turns

Figure 2 shows mean retrieval relevance scores (TF-IDF cosine similarity) across turns 1–5. We observe a modest decline. Mean relevance drops from **0.323 at turn 1** to **0.301 at turn 4** — a **6.6% relative decline** (unpaired t-test, $p = 0.046$) — and is 0.278 at turn 5 (3 sessions). A regression of similarity on turn number over all 550 turns, with standard errors clustered by session, gives a slope of -0.0075 per turn (95% CI $[-0.0126, -0.0024]$, $p = 0.004$). Part of the pooled difference reflects which sessions reach later turns: sessions that reach turn 4 had lower turn-1 similarity than those that stop earlier (0.311 vs. 0.353, Welch $p = 0.033$), and within those 108 sessions the change from turn 1 to turn 4 is not significant (paired $p = 0.17$). Because the retrieval query for turn i contains only the original question and q_i, this decline must come from the sub-queries, the index, or the sessions that reach later steps, not from accumulated context affecting the retriever.

Turn	N Sessions	Mean Similarity	Mean Tokens
1	150	0.323	482
2	147	0.316	972
3	142	0.307	1,464
4	108	0.301	2,095
5	3	0.278	3,611

Table 2: Per-turn retrieval statistics across 150 sessions.

Figure 2: Retrieval quality across turns

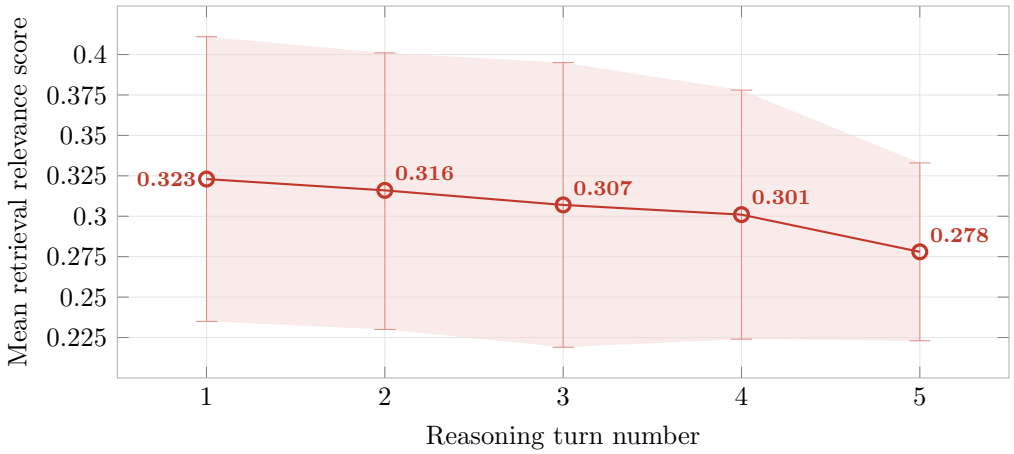


Figure 2: Mean retrieval relevance score (top-chunk TF-IDF cosine similarity) by reasoning turn (0.323 → 0.301 from turn 1 to turn 4, a 6.6% relative decline, unpaired $p = 0.046$; turn 5 has only 3 sessions). Shaded region and error bars show ±1 standard deviation.

We note that the magnitude of degradation (6.6%) is **modest** compared to what might be expected in open-domain settings. This may reflect the constrained mathematical domain or the query anchoring (Section 3.3), which we could not test separately. The decline is gradual rather than consistent at every transition (turn 1 to turn 2: $p = 0.48$).

4.4 Keyword Drift Across Turns

To measure keyword drift, we computed the TF-IDF cosine similarity between consecutive sub-queries of a session, using a single vectorizer fitted on all sub-queries. Mean similarity was 0.257 for turns 1–2 ($n = 147$), 0.247 for turns 2–3 ($n = 142$) and 0.243 for turns 3–4 ($n = 108$), as shown in Figure 3; turns 4–5 have only 3 sessions and are not shown. The values decrease slightly, but a session-clustered regression on the transition index gives a slope of −0.006 per transition ($p = 0.56$), so we find no reliable evidence of keyword drift in these data.

Figure 3: Keyword drift across turns

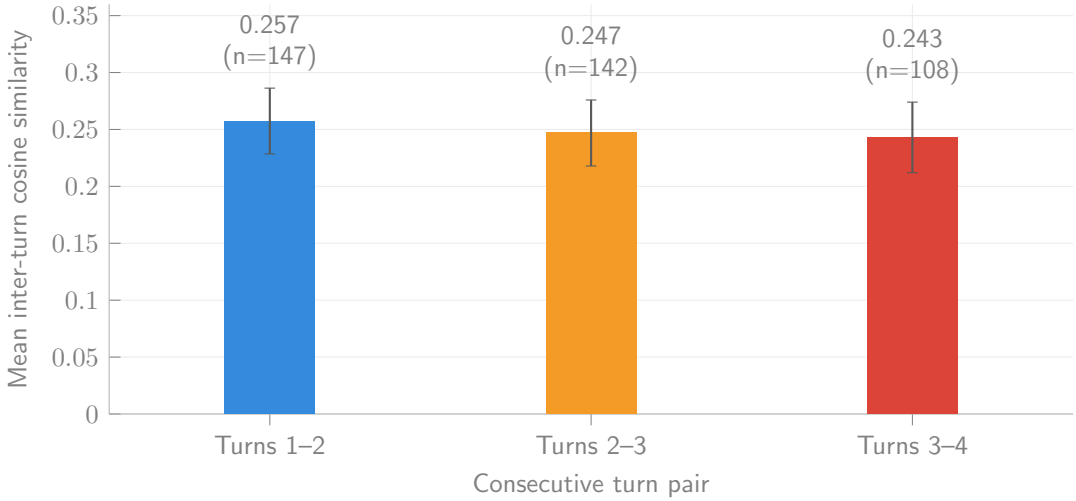


Figure 3: Mean similarity between consecutive sub-queries (turns 1–4), computed with a single TF-IDF vectorizer fitted on all sub-queries. Error bars are 95% CIs. The decrease is not statistically reliable ($p = 0.56$).

4.5 Relationship Between Context Length and Retrieval Quality

Figure 4 shows the scatter relationship between context token count and retrieval relevance across all 547 Tier 1 retrieval turns. The correlation is moderate, negative and statistically significant (Pearson $r = -0.283$, $p < 0.0001$, $n = 547$; Spearman $\rho = -0.295$). A session-level bootstrap gives a 95% CI for r of $[-0.36, -0.20]$. Context length is the strongest of the variables we logged (it explains 8.0% of the variance in similarity, versus 1.2% for turn number), and the association remains after adjusting for turn number, sub-query length and problem type (about -0.04 cosine per 1,000 tokens, session-clustered). However, with session fixed effects, context length and turn number cannot be separated (context coefficient $+0.003$, 95% CI $[-0.017, +0.022]$; turn coefficient $p = 0.23$), so we do not claim that context length itself causes the decline. Since accumulated context does not enter retrieval in this system, context length is best read as a proxy for position in the plan and for problem complexity.

Figure 4: Context length vs retrieval quality

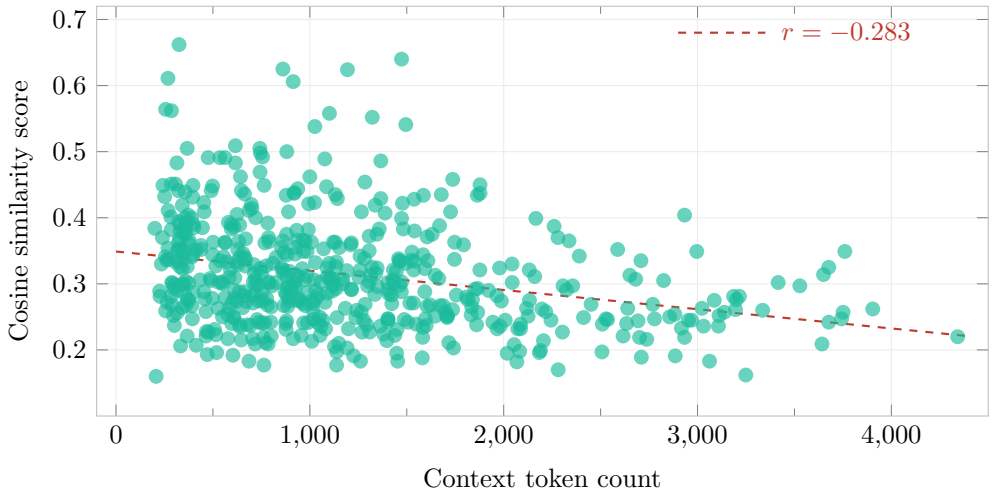


Figure 4: Moderate negative correlation between context token count and retrieval cosine similarity (Pearson $r = -0.283$, $p < 0.0001$, $n = 547$ Tier 1 turns). Each dot is one logged reasoning turn. Dashed line = linear regression.

5. Analysis: Possible Explanations

Because retrieval depends only on the original question, the sub-query and the index, the association between similarity and reasoning length must arise through properties of the sub-queries or of the sessions. We list candidate explanations; our logs do not allow us to distinguish them.

5.1 Sub-query Specificity and Plan Structure

Decomposition tends to order sub-queries from setup towards computation and final simplification. Later sub-queries may therefore be more specific or computational and share fewer terms with the chunks in the index, which lowers TF-IDF similarity. Keyword choice is known to matter for retrieval (Hayashi et al., 2025); here it varies across the steps of a plan.

5.2 Problem Difficulty

Harder problems produce longer reasoning, so long contexts are more common in sessions with demanding questions, whose sub-queries may also match the index less well. Adjusting for problem type and sub-query length does not remove the association between similarity and context length, but session-level difficulty is not measured directly.

5.3 Query Composition and Index Coverage

With anchoring, every retrieval query contains the full original question, which is constant within a session, so similarity depends on how much each sub-query adds to or dilutes it. Similarity also depends on index coverage: scores are computed against a small index whose IDF is recomputed from its own chunks, so they are not comparable across different indices.

5.4 Context Effects in Other Designs

In systems that generate later sub-queries or retrieval queries conditioned on accumulated context, context-length interference could arise, for example through lexical overlap with earlier passages or through attention effects inside the LLM (Liu et al., 2023; Ma and Okazaki, 2026). Our system does not work this way, so our data do not test these mechanisms.

6. Turn-Aware Retrieval Confidence Estimator

6.1 Motivation

Our empirical findings motivate a practical question: can a lightweight signal flag sessions likely to end in an incorrect final answer, without requiring expensive human annotation?

6.2 Estimator Design

We define the confidence score as:

$$\text{conf}(t,\, |C|,\, g) \;=\; g \;\times\; \exp(-\alpha \times t) \;\times\; \big(1 - \beta \times \log(|C| \,/\, C_{\text{baseline}})\big)$$

where t = current reasoning turn number; $|C|$ = current context token count; g = normalised TF-IDF cosine similarity; $C_{\text{baseline}} = 482$ tokens (mean turn-1 context length); $\alpha = 0.18$ (turn-number decay); $\beta = 0.09$ (context length penalty). The score combines raw retrieval similarity, a turn-position discount, and a logarithmic penalty on context length.

6.3 Evaluation

We evaluate the estimator using binary correctness labels from 102 labeled sessions (94 correct, 8 incorrect). Each label is the user's feedback on the final answer, stored on the session's last turn, and is paired with that turn's index, context length and similarity. We note that the minority class contains only 8 incorrect labels; results should be interpreted as **preliminary and directional** rather than definitive. Table 3 reports AUROC with percentile bootstrap 95% confidence intervals (2,000 resamples of turns).

Method	AUROC	95% CI
Random baseline	0.500	—
Turn number only	0.563	[0.30, 0.80]
Grounding similarity only	0.553	[0.37, 0.74]
Turn-aware estimator (ours)	0.634	[0.43, 0.84]
Context length only	0.695	[0.42, 0.92]

Table 3: Detection of incorrect final answers on labeled sessions ($N = 102$ sessions). 8 incorrect labels; all intervals include 0.5.

The turn-aware estimator achieves AUROC = **0.634**. Context length alone has a higher point estimate (**AUROC = 0.695**), but all intervals include 0.5 and the 95% CI for the difference between context length and turn number is $[-0.04, +0.28]$. Labels are concentrated at turn 4 (78 of 102 sessions), which leaves little turn variation for the turn-only baseline; because labels sit on the last turn, these features describe session length and final context size rather than the quality of any single retrieval. We regard these results as exploratory.

Figure 5: Confidence estimator ROC curve

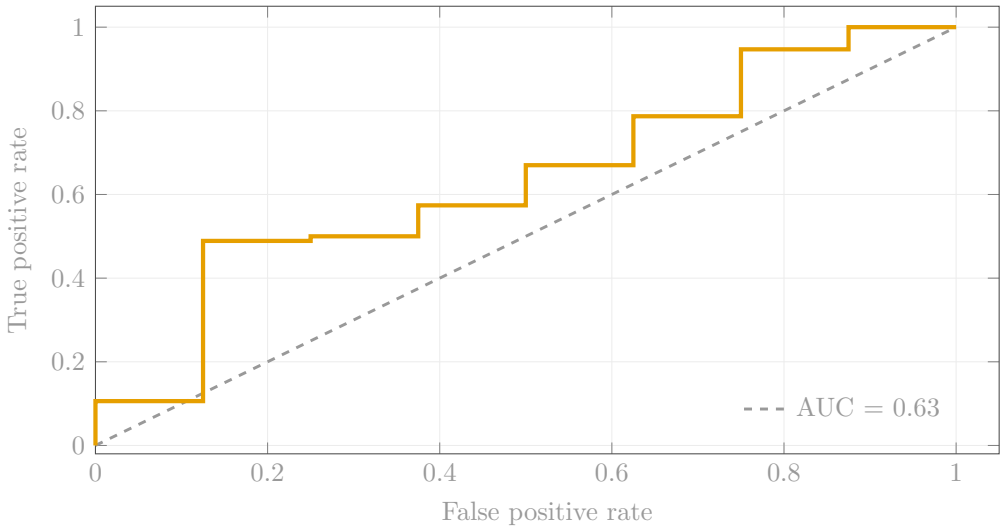


Figure 5: ROC curve for the turn-aware confidence estimator (AUC = 0.634) on the 102 labeled sessions (94 correct, 8 incorrect); each label is user feedback on the final answer. The dashed diagonal is the random baseline (AUC = 0.500).

7. Discussion

7.1 Query Anchoring

Among the sessions with feedback, 94 of 102 final answers were marked correct despite the measured decline in similarity. Retrieving with the original question plus the sub-query may help keep retrieval on topic, but we cannot test this because anchoring was enabled in every logged turn, and only 68% of sessions gave feedback.

7.2 Implications for RAG Evaluation

Many existing benchmarks assume single-turn retrieval and do not examine how retrieval varies across the steps of a session. We argue that agentic RAG evaluation requires: (1) multi-turn sessions with at least 3 reasoning turns; (2) retrieval quality measurement at each turn independently; (3) per-step retrieval trends reported as a metric; and (4) logging of exactly what the retriever receives at each turn, so that effects of plan structure can be separated from effects of context.

7.3 Implications for System Design

Per-step retrieval evaluation. Since later steps may retrieve less well for reasons unrelated to context, retrieval should be evaluated per step, with relevance judgments, before design changes such as pruning context are proposed.

Turn-aware keyword generation. Sub-query generation could re-anchor explicitly to the original input query at each turn; whether this helps is untested here.

Retrieval confidence monitoring. A simple context length threshold (AUROC = 0.695, wide confidence interval) may be worth evaluating further.

7.4 Limitations

Closed-source LLM. Our deployment uses Gemini 2.5 Pro, so we cannot analyze attention weights.

Single system and domain. The data come from one system and one domain (mathematics), collected at one site over three days (150 sessions, 145 distinct questions), where multi-step structure may produce specific patterns.

Similarity is not relevance. TF-IDF similarity scores are not relevance judgments, and we have no labels of retrieval relevance.

Index. The index was a small, researcher-prepared set of chunks with IDF recomputed from it, so scores are not comparable across indices, and the index could change during collection if refined answers were re-indexed.

Retriever independence from context. Retrieval queries do not contain accumulated context, so our data do not test context effects on the retriever. The context token count is approximate and is measured after each step.

Small effect and inseparable variables. The decline in similarity is small, and part of the pooled turn 1 to turn 4 difference reflects which sessions reach later turns. Context length and turn number are inseparable within sessions, so our results are associations, not causal effects.

Anchoring and labels. Query anchoring was enabled in every logged turn, so its effect is untested. Correctness labels are users' feedback on final answers for 68% of sessions and include only 8 errors, so they are self-selected and all AUROC estimates are imprecise; C_{baseline} and the estimator parameters were set on the same data used for evaluation.

8. Conclusion

We have presented an empirical study of multi-turn retrieval degradation, defined as declining retrieval similarity across reasoning turns, in a production agentic RAG system. Across 150 sessions (550 turn observations) we document: (1) a modest decline in retrieval similarity across turns (-0.0075 per turn, $p = 0.004$, session-clustered; 6.6% between turn 1 and turn 4, $p = 0.046$);

(2) a moderate negative correlation between accumulated reasoning length and retrieval similarity ($r = -0.283$, $p < 0.0001$); and (3) no reliable evidence of keyword drift between consecutive sub-queries.

Because the retrieval query never contains accumulated context, and context length and turn number are inseparable within sessions, we attribute the decline to the sub-queries, the index or the sessions that reach later steps, rather than to context interference. Context length has the highest point estimate as a detector of incorrect final answers (AUROC = 0.695), but with wide confidence intervals. These findings suggest that single-turn RAG benchmarks should be supplemented with per-step evaluation, and that controlled experiments, dense-retriever replication and relevance labels are needed to establish how retrieval varies across the steps of agentic sessions.

References

Vaibhav Adlakha, Shehzaad Dhuliawala, Kaheer Suleman, Harm de Vries, and Siva Reddy (2022). TopiOCQA: Open-domain conversational question answering with topic switching. *TACL 10*, 468–483.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi (2023). Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR 2024*.

Kazuki Hayashi, Hidetaka Kamigaito, Shinya Kouda, and Taro Watanabe (2025). IterKey: Iterative keyword generation with LLMs for enhanced retrieval augmented generation. *COLM 2025*.

Zhengbao Jiang, Frank F. Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig (2023). Active retrieval augmented generation. *EMNLP 2023*.

Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih (2020). Dense passage retrieval for open-domain question answering. *EMNLP 2020*.

Yannis Katsis, Sara Rosenthal, Kshitij Fadnis, Chulaka Gunasekara, Young-Suk Lee, Lucian Popa, Vraj Shah, Huaiyu Zhu, Danish Contractor, and Marina Danilevsky (2025). MTRAG: A multi-turn conversational benchmark for evaluating retrieval-augmented generation systems. *TACL 13*, 784–808.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS 33*, 9459–9474.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang (2023). Lost in the middle: How language models use long contexts. *TACL 12*, 157–173.

Youmi Ma and Naoaki Okazaki (2026). From interpretability to performance: Optimizing retrieval heads for long-context language models. *arXiv:2601.11020*.

Shintaro Ozaki, Yuta Kato, Siyuan Feng, Masayo Tomita, Kazuki Hayashi, Wataru Hashimoto, Ryoma Obara, Masafumi Oyamada, Katsuhiko Hayashi, Hidetaka Kamigaito, and Taro Watanabe (2025). Understanding the impact of confidence in retrieval augmented generation: A case study in the medical domain. *BioNLP 2025*.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling (2024). Corrective retrieval augmented generation. *arXiv:2401.15884*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan

Cao (2023). ReAct: Synergizing reasoning and acting in language models. *ICLR 2023*.

Hiyori Yoshikawa and Naoaki Okazaki (2023). Selective-LAMA: Selective prediction for confidence-aware evaluation of language models. *Findings of EACL 2023*.

Appendix A. Logging Implementation

The per-turn logging system was implemented as a TypeScript service (`logger.ts`) integrated into the application's reasoning loop. Each reasoning turn logs the following fields once its reasoning step has completed:

```
interface TurnLog {
  sessionId: string           // Random session identifier
  turnNumber: number          // 1-indexed turn within session
  subQuery: string            // Sub-query text
  originalQuery: string       // Original user question
  ragChunkCount: number       // Chunks retrieved by TF-IDF
  groundingChunkCount: number // Web-grounding chunks
  contextTokenCount: number   // Approx. tokens (chars/4) of
                              // accumulated context, after the step
  usedWebSearch: boolean      // Whether Google Search was used
  retrievalTier: number       // 1=TF-IDF, 2=KB, 3=Web
  similarityScore: number     // Top chunk cosine similarity
  anchoringEnabled: boolean   // Whether query anchoring was active
  problemType: string         // algebra/calculus/geometry/statistics
  timestamp: string           // ISO 8601 timestamp
  wasCorrect?: boolean        // Final-answer feedback, on last turn
}
```

Logs are stored in browser localStorage under key `rag_research_logs` and exported as JSON via a download interface. Session boundaries are determined by new question submission — each question initiates a new `sessionId` automatically via `resetSession()`.

Appendix B. Parameter Details

Parameters $\alpha = 0.18$ and $\beta = 0.09$ in the confidence estimator were selected based on the assumed relationship between turn number decay and context length penalty in our system. $C_{\text{baseline}} = 482$ tokens represents the empirically measured mean turn-1 context length across all 150 sessions. The parameters were not cross-validated, and C_{baseline} was computed on the same data used for evaluation.

Appendix C. Statistical Details

All regressions are ordinary least squares with cluster-robust standard errors clustered by session, with t critical values on $G - 1 = 149$ degrees of freedom. Within-session specifications subtract each session's mean from the outcome and predictors. The bootstrap CI for the correlation resamples sessions (2,000 resamples); AUROC CIs resample turns (2,000 resamples).